

GRAND AURELIA

HOTEL & RESORT • EST. 1928 — ENTERPRISE HOSPITALITY PLATFORM

Where Timeless Luxury Meets Culinary Artistry & Intelligent Technology

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<https://grand-aurelia-five.vercel.app/>

1. Executive Summary & Vision

Grand Aurelia is an enterprise-grade, full-stack digital management ecosystem tailored for luxury 5-star hotels, fine-dining restaurants, rapid cloud kitchens, and gourmet food delivery operations. The platform seamlessly unifies multi-department hospitality workflows into a single high-performance interface with sub-millisecond responsiveness, real-time synchronization, and a 24/7 AI Concierge.

React 19

Vite 8

Node.js & Express

REST API Engine

24/7 AI Concierge

Vercel Cloud Live

2. Core Enterprise Pillars

- **Role-Based Access Control (RBAC):** Simulates 7 operational roles: General Manager, Front Desk, Executive Chef, Waiter POS, Delivery Rider, Housekeeping Lead, and Public Guest.
- **Hotel Accommodations & Front Desk:** Real-time room availability matrix, 1-click guest check-in/checkout, automated housekeeping turnover queuing, and in-room dining integration.
- **Fine Dining & Table POS:** Live floor plan (Main Dining, VIP Lounge, Terrace), waiter POS order dispatching directly to the kitchen, reservations, and instant bill settlement.
- **Kitchen Display System (KDS):** Live kitchen ticket management with preparation countdown timers and stage-by-stage status advancement.
- **Gourmet Food Delivery Hub:** Interactive digital menu, smart promo discount calculations, multi-payment gateway (bKash/Nagad/Cards/COD), and 4-step live rider tracking.
- **Housekeeping Turnover Kanban:** 3-stage room cleaning board with automatic front-desk room availability synchronization.
- **Unified Stock & Inventory:** Cross-department supplies monitoring with auto low-stock threshold warnings and 1-click restock.
- **Digital Folios & Invoicing Hub:** Consolidated billing covering room nights, restaurant meals, taxes, and printable official receipts.
- **24/7 AI Smart Concierge:** Intelligent hospitality bot answering guest inquiries, suggesting room upgrades, and triggering instant UI actions.

Grand Aurelia — Detailed Module Breakdown

Departmental features, business workflows, and interactive capabilities

3. Comprehensive Module Specifications

3.1 Executive GM Overview Dashboard

The executive command center provides leadership with real-time visibility into hospitality operations:

- **KPI Performance Cards:** Tracks Total Platform Revenue, Hotel Occupancy Rate, Active Dine & Delivery Orders, and Critical Low-Stock Alerts with matching brand borders and smooth hover elevation.
- **Revenue Contribution Split:** Visual progress bars analyzing accommodation earnings vs culinary & delivery sales.
- **Live Operations Activity Stream:** Real-time feed of recent guest bookings, check-ins, and incoming culinary orders.

3.2 Hotel Accommodations & Front Desk Operations

Manages luxury suite inventories across Presidential Penthouses, Royal Suites, Executive Balcony, and Deluxe Garden rooms:

- **Availability Grid:** Color-coded room statuses: Available (Green), Occupied (Blue), Cleaning (Amber), Maintenance (Red).
- **Instant Booking Modal:** Collects guest name, contact, stay dates, guest count, and calculates total cost with tax.
- **1-Click Check-In & Check-Out:** Streamlines guest arrivals and automatically triggers cleaning tasks upon checkout.

3.3 Fine Dining Table POS & Table Management

Designed for floor captains and waiters to manage reservations and table orders:

- **Interactive Floor Grid:** Displays table capacities, current status (Available, Occupied, Reserved), and active orders.
- **Waiter POS Dispatch:** Waiters can select dishes from the culinary catalog and fire tickets straight to Kitchen KDS.
- **Table Reservation Engine:** Enables advance table booking with special requests (Anniversary, Window seat, Business dinner).

3.4 Real-Time Kitchen Display System (KDS)

Replaces paper tickets with a synchronized kitchen display screen:

- **Ticket Stages:** Order Placed -> Preparing / Cooking -> Ready for Table / Hand to Rider -> Served / Dispatched.
- **Order Timers & Urgency:** Visual color indicators showing order age to ensure compliance with fine-dining SLAs.

Grand Aurelia — Operations, Logistics & AI

Online delivery, facility turnover, supply chain, and intelligent agent

3.5 Gourmet Express Food Delivery & Rider App

Public-facing online food delivery and dedicated rider dispatch interface:

- **Menu Catalog & Cart:** Filter by Chef Specials, Main Courses, Appetizers, Desserts, and Beverages with animated counters.
- **Checkout & Payment:** Supports bKash, Nagad, Credit/Debit Cards, and Cash on Delivery with delivery address validation.
- **Live 4-Stage Rider Tracking:** Simulates live GPS rider route, vehicle details, contact button, and delivery confirmation.

3.6 Housekeeping & Facility Turnover Management

Ensures 5-star sanitization and cleanliness standards across all hotel suites:

- **Kanban Board:** 3-column drag/click workflow: Pending Cleaning -> In-Progress -> Cleaned & Inspected.
- **Automated Room Sync:** Marking a task "Cleaned" automatically switches room status on Front Desk to "Available".

3.7 Unified Stock & Multi-Department Inventory

Centralized warehouse tracking kitchen raw ingredients, bar beverages, and room amenities:

- **Threshold Alert System:** Visual warning badges when items fall below safe threshold limits.
- **1-Click Restocking:** Instant restock buttons to replenish quantities with real-time database update.

3.8 Digital Billing, Folios & Invoicing Hub

Consolidated guest billing management:

- **Guest Folios:** Consolidates room stay charges, dining expenses, service fees, and taxes into itemized statements.
- **Printable Receipts:** Branded 5-star official receipts with 1-click browser print integration.

3.9 24/7 AI Smart Concierge & Knowledge Engine

Autonomous conversational agent trained on Grand Aurelia hotel policies and menus:

- **Automated Recommendations:** Recommends suites based on guest party size and suggests wine/dining pairings.
- **Quick Action Chips:** Allows guests to instantly jump to suites, order food, or reserve dining tables with 1 click.

Grand Aurelia — Technical Architecture & Deployment

Codebase structure, REST endpoints, deployment, and handover details

4. Technical Architecture & File Structure

The codebase follows clean architecture separating frontend presentation from backend business logic:

```
Nasir Vai_project/
|-- client/                # React 19 + Vite 8 Frontend Application
|   |-- src/components/    # GMDashboard, HotelView, RestaurantView, KDSView, etc.
|   |-- src/App.jsx        # Main Application Root State & Safe Sync Engine
|   +-- src/index.css      # Grand Aurelia Luxury Design Tokens (#1F3A5F & #FF2147)
|-- server/               # Node.js + Express.js REST API Backend
|   |-- data/database.json # Persistent JSON Database Seed
|   +-- src/index.js       # REST API Routes, Middleware & CRUD Handlers
+-- start.bat             # 1-Click Windows Production / Dev Launcher
```

5. REST API Endpoints Specification

GET / POST	/api/rooms	Retrieve all rooms & create room booking
POST	/api/rooms/:id/checkin	Perform 1-click guest check-in
POST	/api/rooms/:id/checkout	Check-out guest & dispatch cleaning
GET / POST	/api/tables	Get floor tables & reserve table
GET / POST	/api/orders	Fetch culinary orders & fire new tickets
PATCH	/api/orders/:id/status	Update KDS cooking / delivery status
GET / PATCH	/api/housekeeping	Manage room cleaning Kanban tasks
GET / POST	/api/inventory	Monitor stock levels & execute restock
GET	/api/analytics	Consolidated revenue & occupancy metrics
GET	/api/health	Production uptime & health check endpoint

6. Deployment & Developer Credits

- **Live Production Cloud URL:** <https://grand-aurelia-five.vercel.app/>
- **GitHub Official Repository:** <https://github.com/rokeyaag/grand-aurelia>
- **Local Launch Instruction:** Double-click start.bat in the root folder or execute "npm start".
- **Lead Software Developer:** Lutfor Rahman
- **Engineering Organization:** Lutfor IT Solution
- **Copyright & License:** © 2026 Grand Aurelia Luxury Hotel & Resort. All rights reserved. MIT License.